Siyang Li

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RESEARCH

- Distances to nearby galaxies and the Hubble constant with three independent distance indicators: Cepheids, the tip of the red giant branch (TRGB), and the J-region asymptotic giant branch (JAGB).
- PI of the CHASE spectral survey, to standardize the JAGB as a distance indicator independent of Cepheids & TRGB.
- Lead organizer of the CosmoVerse Data Challenge: a transparent, community-wide test of how TRGB methods affect H_0, with public data, pipelines, and results, and materials accessible without prior expertise.

APPOINTMENTS

Brinson Prize Postdoctoral Fellow, UC Berkeley (Host: Alexei V. Filippenko) 2025–present

FELLOWSHIPS

Brinson Prize Postdoctoral Fellowship ($360,000) 2025–2028

International Space Science Institute (ISSI) Postdoctoral Fellowship (declined) 2025

National Science Foundation Graduate Research Fellowship ($108,000) 2022–2025

TELESCOPE TIME

PI, Lick Observatory Shane 3-meter Telescope — 3 nights each semester 2026A, 2026B
Refining a New Hubble Tension Probe: Testing J-region Asymptotic Giant Branch Homogeneity Assumptions

PI, Apache Point Observatory ARC 3.5-meter Telescope — 7 half-nights 2024–2025
Towards Standardization of the J-region Asymptotic Giant Branch: Investigating Asymmetric Luminosity Functions with Carbon Star Spectra

PI, Hubble Space Telescope Cycle 32 Bridge, GO-17911 — 8 orbits ($25,000) 2024
Building the Last Bridge Between the PHAT & PHATTER Surveys and Carbon Star Cosmology

Co-I, Hubble Space Telescope Cycle 31, GO-17520 (PI: L. Breuval) — 42 orbits 2023
A 1% cross-calibration of Cepheids, TRGB, and JAGB in five nearby galaxies with HST

EDUCATION

Johns Hopkins University 2020–2025
Doctor of Philosophy in Astronomy & Astrophysics
Master of Arts in Physics 2020–2023
Dissertation: *Testing Hubble Tension Systematics using Cepheids, Tip of the Red Giant Branch, and J-region Asymptotic Giant Branch*
Thesis Advisor: Professor Adam G. Riess

University of California, Berkeley 2016–2020
Bachelor of Arts, High Honors in Physics, cum laude
Honors Thesis: *Characterization of a High Efficiency Silicon Photomultiplier for Astrophysical Transient Searches*
Thesis Advisor: Professor George F. Smoot III

FIRST-AUTHOR PUBLICATIONS

Citations to all 13 first-author publications (8 refereed, 1 submitted, 4 proceedings) as of October 16, 2026: 248.
†Student mentored by S. Li.

9. **Siyang Li**, Alexei V. Filippenko, Arief Ahmad, Adam G. Riess, Stefano Casertano, Thomas G. Brink, Richard I. Anderson, Caroline D. Huang, Louise Breuval, Noriyuki Matsunaga, Gracelynn Jost†, Yash Mehta†, Elaina Sadler†, Julian Shapiro†. "CHASE Spectral Survey I: Measuring Contamination in the Color-selected J-region Asymptotic Giant Branch Distance Indicator." *Submitted to ApJ* (2026). ADS
8. **Siyang Li**, Adam G. Riess, Gagandeep S. Anand, Daniel Scolnic, Yukei S. Murakami, Dillon Brout, Erik R. Peterson. "The Complete Sample of Available SNe Ia Luminosity Calibrations from the TRGB Observed with either *HST* or *JWST*." *ApJ*, 997, 115 (2026). ADS
7. **Siyang Li**, Adam G. Riess, Daniel Scolnic, Stefano Casertano, Gagandeep S. Anand. "JAGB 2.0: Improved Constraints on the J-region Asymptotic Giant Branch-based Hubble Constant from an Expanded Sample of *JWST* Observations." *ApJ*, 988, 97 (2025). ADS
6. **Siyang Li**, Gagandeep S. Anand, Adam G. Riess, Stefano Casertano, Wenlong Yuan, Louise Breuval, Lucas M. Macri, Daniel Scolnic, Rachael Beaton, Richard I. Anderson. "Tip of the Red Giant Branch Distances with *JWST*. II. I-band Measurements in a Sample of Hosts of 10 Type Ia Supernova Match *HST* Cepheids." *ApJ*, 976, 177 (2024). ADS
5. **Siyang Li**, Adam G. Riess, Stefano Casertano, Gagandeep S. Anand, Daniel Scolnic, Wenlong Yuan, Louise Breuval, Caroline D. Huang. "Reconnaissance with *JWST* of the J-region Asymptotic Giant Branch in Distance Ladder Galaxies: From Irregular Luminosity Functions to Approximation of the Hubble Constant." *ApJ*, 966, 20 (2024). ADS
4. **Siyang Li**, Adam G. Riess, Daniel Scolnic, Gagandeep S. Anand, Jiaxi Wu, Stefano Casertano, Wenlong Yuan, Rachael Beaton, Richard I. Anderson. "Standardized Luminosity of the Tip of the Red Giant Branch utilizing Multiple Fields in NGC 4258 and the CATs Algorithm." *ApJ*, 956, 32 (2023). ADS
3. **Siyang Li**, Stefano Casertano, Adam G. Riess. "A *Gaia* Data Release 3 View on the Tip of the Red Giant Branch Luminosity." *ApJ*, 950, 83 (2023). ADS
2. **Siyang Li**, Stefano Casertano, Adam G. Riess. "A Maximum Likelihood Calibration of the Tip of the Red Giant Branch Luminosity from High Latitude Field Giants using *Gaia* Early Data Release 3 Parallaxes." *ApJ*, 939, 96 (2022). ADS
1. **Siyang Li**, Adam G. Riess, Michael P. Busch, Stefano Casertano, Lucas M. Macri, Wenlong Yuan. "A Sub-2% Distance to M31 from Photometrically Homogeneous Near-infrared Cepheid Period-Luminosity Relations Measured with the *Hubble Space Telescope*." *ApJ*, 920, 84 (2021). ADS

FIRST-AUTHOR CONFERENCE PROCEEDINGS

4. **Siyang Li**, George F. Smoot III. "Characterization of a high efficiency silicon photomultiplier for millisecond to sub-microsecond astrophysical transient searches." *SPIE Proc.*, 11454, 1145422 (2020). ADS
3. **Siyang Li**, George F. Smoot III, Bruce Grossan, Albert Wai Kit Lau, Marzhan Bekbalanova, Mehdi Shafiee, Thorsten Stezelberger. "Program objectives and specifications for the Ultra-Fast Astronomy observatory." *SPIE Proc.*, 11341, 113411Y (2019). ADS
2. **Siyang Li**, George F. Smoot III. "Characterization of a silicon photomultiplier for the Ultra-Fast Astronomy telescope." *SPIE Proc.*, 11115, 111150A (2019). ADS
1. **Siyang Li**, Jérôme Maire, Maren Cosens, Shelley A. Wright. "Detector characterization of a near-infrared discrete avalanche photodiode 5×5 array for astrophysical observations." *SPIE Proc.*, 11002, 110022G (2019). ADS

CO-AUTHOR PUBLICATIONS

Total citations as of October 16, 2026: 1052 (531 excluding the CosmoVerse White Paper)

10. Stefano Casertano & the H_0 Distance Network Collaboration including **Siyang Li**. "The Local Distance Network: A community consensus report on the measurement of the Hubble constant at ~1% precision." *A&A*, 708, A166 (2026). ADS
9. Adam G. Riess, **Siyang Li**, Gagandeep S. Anand, Wenlong Yuan, Louise Breuval, Stefano Casertano, Lucas M. Macri, Daniel Scolnic, Yukei S. Murakami, Alexei V. Filippenko, Thomas G. Brink. "The Perfect Host: *JWST* Cepheid Observations in a Background-free Type Ia Supernova Host Confirm No Bias in Hubble-constant Measurements." *ApJL*, 992, L34 (2025). ADS
8. Eleonora Di Valentino et al. including **Siyang Li**. "The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics." *PDU*, 49, 101965 (2025). ADS
   - *J-region Asymptotic Giant Branch* — Principal Author: **Siyang Li**; Co-authors: A. Riess, S. Casertano, G. Sloan, L. Breuval, B. Zgirski, G. Anand, R. Anderson, C. Huang, D. Scolnic.
   - *Tip of the Red Giant Branch* — Principal Author: R. I. Anderson; Co-authors: G. S. Anand, I. Saltas, G. De Somma, **S. Li**, V. Vansevicius, L. Breuval, A. Riess.
   - *Cepheids* — Principal Author: L. Breuval; Co-authors: R. I. Anderson, G. De Somma, L. Galbany, **S. Li**, L. Macri, L. Perivolaropoulos, A. Riess, V. Vansevicius.
7. Rolf-Peter Kudritzki, Miguel A. Urbaneja, Fabio Bresolin, Lucas M. Macri, Wenlong Yuan, **Siyang Li**, Gagandeep S. Anand, Adam G. Riess. "The Hubble Constant Anchor Galaxy NGC 4258: Metallicity and Distance from Blue Supergiants." *ApJ*, 977, 217 (2024). ADS
6. Adam G. Riess, Daniel Scolnic, Gagandeep S. Anand, Louise Breuval, Stefano Casertano, Lucas M. Macri, **Siyang Li**, et al. "*JWST* Validates *HST* Distance Measurements: Selection of Supernova Subsample Explains Differences in *JWST* Estimates of Local H_0." *ApJ*, 977, 120 (2024). ADS
5. Gagandeep S. Anand, Adam G. Riess, Wenlong Yuan, Rachael Beaton, Stefano Casertano, **Siyang Li**, et al. "Tip of the Red Giant Branch Distances with *JWST*: An Absolute Calibration in NGC 4258 and First Applications to Type Ia Supernova Hosts." *ApJ*, 966, 89 (2024). ADS
4. Daniel Scolnic, Adam G. Riess, Jiaxi Wu, **Siyang Li**, Gagandeep S. Anand, Rachael Beaton, Stefano Casertano, Richard I. Anderson, Suhail Dhawan, Xinwei Ke. "CATS: The Hubble Constant from Standardized TRGB and Type Ia Supernova Measurements." *ApJL*, 954, L31 (2023). ADS
3. Louise Breuval, Adam G. Riess, Lucas M. Macri, **Siyang Li**, Wenlong Yuan, Stefano Casertano, Tarini Konchady, Boris Trahin, Meredith J. Durbin, Benjamin F. Williams. "A 1.3% Distance to M33 from *Hubble Space Telescope* Cepheid Photometry." *ApJ*, 951, 118 (2023). ADS
2. Jiaxi Wu, Daniel Scolnic, Adam G. Riess, Gagandeep S. Anand, Rachael Beaton, Stefano Casertano, Xinwei Ke, **Siyang Li**. "Comparative Analysis of TRGBs (CATs) from Unsupervised, Multi-Halo-Field Measurements: Contrast is Key." *ApJ*, 954, 87 (2023). ADS
1. Albert W. K. Lau, Mehdi Shafiee, George F. Smoot III, Bruce Grossan, **Siyang Li**, Zhanat Maksut. "On-sky silicon photomultiplier detector performance measurements for millisecond to sub-microsecond optical source variability studies." *JATIS*, 6(4), 046002 (2020). ADS

BOOK CHAPTER AND PATENT

INVITED BOOK CHAPTER

Siyang Li, Rachael L. Beaton. "Tip of the Red Giant Branch." Invited chapter in *The Hubble Constant Tension* (Eds. E. Di Valentino & D. Brout), Springer Singapore, 2024. DOI: 10.1007/978-981-99-0177-7_8.

PATENT

Siyang Li, George F. Smoot III, Thorsten Stezelberger. "Silicon Photomultiplier Telescope Camera." U.S. Provisional Patent Application No. 62/940,250, filed November 26, 2019 (provisional only; no full application filed).

HONORS AND AWARDS

HONORS

American Astronomical Society Rodger Doxsey Travel Prize 2024
One of 8 recipients selected from 47 entries; funded travel to AAS 245

Lindau Nobel Laureate Meeting (Physics), selected participant 2024
73rd meeting, Lindau, Germany

UNDERGRADUATE AWARDS

UC Berkeley Regents' and Chancellor's Scholarship 2016–2020
Awarded to the top 2% of undergraduates based on academic and extracurricular achievements

Hong Kong University of Science and Technology International Research Fellowship 2017–2019
Fully covers airfare, accommodation, and meals for research visits to HKUST

SPIE Optics and Photonics Education Scholarship 2019
One of 84 awarded worldwide, for potential contribution to optics and photonics

Other undergraduate research and travel awards: UC Berkeley Summer Undergraduate Research Fellowship, Physics Undergraduate Research Scholarship, Regents' and Chancellor's Research Fellowship, and Academic Opportunity Fund Grant; Measurement Science Conference Scholarship; Best Poster Award at the 8th Applied Optics and Photonics China Conference 2017–2020

TALKS

INVITED TALKS

Galileo Galilei Institute Exploring New Frontiers in Cosmology, Florence, Italy (virtual) Jul 2026

Cross-Survey Cosmography Conference, New Delhi, India Apr 2026

33rd Texas Symposium on Relativistic Astrophysics (memorial of G. F. Smoot III), Tempe, AZ Dec 2025

Brinson Prize Fellowship Workshop, Chicago, IL Nov 2025

Tensions in Cosmology Conference, Corfu, Greece Sep 2025

NAOJ Cosmology Seminar, jointly with La Sapienza, Tokyo, Japan Jul 2025

University of Tokyo Astronomy Department Colloquium, Tokyo, Japan Jul 2025

Tohoku University Astronomy Department Colloquium, Sendai, Japan Jun 2025

Sauverny Observatory (EPFL) Tea Time Talk, Versoix, Switzerland May 2025

CosmoVerse Workshop@Naples, Naples, Italy May 2025

ISSI Workshop: Towards Consensus on the Local Hubble Constant, Bern, Switzerland — Mar 2025

University of California, Davis Physics Department Seminar, Davis, CA — Jan 2025

Yale University Cosmology Seminar, New Haven, CT — Nov 2024

CosmoVerse Seminar (virtual) — May 2024

Space Telescope Science Institute JWST Science Conversation, Baltimore, MD — Mar 2024

ISSI Annual Meeting: The Stellar Path to the H_0 Tension in the Gaia, TESS, LSST and JWST Era, Bern, Switzerland (virtual) — Jun 2023

CONTRIBUTED TALKS

JASMINE Consortium Workshop: Dynamics and Evolution of the Galactic Center, Mitaka, Japan — Apr 2026

sys2025: Systematic and Measurement Errors across the Sciences, Huntsville, AL — Nov 2025

JASMINE Consortium Meeting, Sendai, Japan — Aug 2025

Kyoto University Department of Astronomy Seminar, Kyoto, Japan — Jul 2025

Kavli IPMU APEC Seminar, Tokyo, Japan — Jul 2025

American Astronomical Society Meeting 245 Cosmology I, National Harbor, MD — Jan 2025

Cosmo'24, Kyoto, Japan — Oct 2024

Resolved Stellar Populations 2024 (Celebration of Peter Stetson's Career), Florence, Italy — Oct 2024

CosmoVerse@Kraków 2024, Kraków, Poland — Jul 2024

The 6th Neighborhood Workshop at Penn State University, State College, PA — Apr 2024

American Astronomical Society Meeting 243 Cosmic Distance Scale II, New Orleans, LA — Jan 2024

The 5th Neighborhood Workshop at Penn State University, State College, PA — Apr 2023

American Astronomical Society Meeting 241 Cosmology II, Seattle, WA — Jan 2023

iid2022 Workshop: Statistical Methods for Event Data, Guntersville, AL — Nov 2022

American Astronomical Society Meeting 240 Cosmology III, Pasadena, CA — Jun 2022

SPIE Astronomical Telescopes + Instrumentation, Conference 11454 (virtual) — Dec 2020

SPIE Optics + Photonics, Conference 11115, San Diego, CA — Aug 2019

Exploring the Energetic Universe Conference, Nur-Sultan, Kazakhstan — Jun 2019

Plus internal seminars at Johns Hopkins University and UC Berkeley — 2019–2025

POSTERS

Cambridge Workshop on Cool Stars, Stellar Systems and the Sun (Cool Stars 23), Tokyo, Japan — Jun 2026

Statistical Challenges in Modern Astronomy VIII, State College, PA — Jun 2023

8th Applied Optics and Photonics China, Conference 10, Beijing, China — Jul 2019

SPIE Defense + Commercial Sensing, Conference 11002, Baltimore, MD — Apr 2019

TEACHING

Lecturer, CosmoVerse Data Challenge School (77 registrants, 31 countries): 5 TRGB lectures — Summer 2026

Teaching Assistant, Stars and the Universe: Cosmic Evolution, Johns Hopkins University — Spring 2021, 2022

Teaching Assistant, General Physics: Physical Science Majors II, Johns Hopkins University — Fall 2020

Teaching Assistant, General Physics Laboratory I, Johns Hopkins University — Fall 2020

MENTORSHIP

Gracelynn Jost, undergraduate student, UC Berkeley; co-author, Li et al. (submitted) 2026
JAGB measurement systematics in the LMC; CosmoVerse Data Challenge

Yash Mehta, undergraduate student, UC Berkeley; co-author, Li et al. (submitted) 2026
Field-to-field variation and standardization of the JAGB in M31; CosmoVerse Data Challenge

Elaina Sadler, undergraduate student, UC Berkeley; co-author, Li et al. (submitted) 2026
Where the JAGB breaks down, by galaxy type and region morphology; CosmoVerse Data Challenge

Julian Shapiro, undergraduate student, UC Berkeley; co-author, Li et al. (submitted) 2026
Planetary nebula luminosity function distance to M101

Gurnoor Singh Sandhu, undergraduate student, UC Berkeley 2026
Alternative methods for fitting the TRGB; CosmoVerse Data Challenge

Dhiraj Kuniyal, PhD student, University of Delhi 2026
Maximum-likelihood TRGB variants analysis

Ryan Lessing, undergraduate student, Johns Hopkins University 2023–2024
Academic and career mentoring through the Physics & Astronomy department program

SERVICE

Grant Proposal Referee, PRELUDIUM program, National Science Centre, Poland 2026

Lead Organizer, CosmoVerse Data Challenge & School 2025–2026

Member, UC Berkeley Astronomy Department Spring Colloquium Committee 2025

Referee, *Scientific Reports* (Springer Nature) 2025

Member, Johns Hopkins Physics & Astronomy Mentorship Committee 2024–2025

Referee, *The Astrophysical Journal Supplement Series* (IOP Publishing) 2024

OUTREACH

First Light Astronomy, public outreach YouTube channel (creator) 2026
Videos on the Hubble tension and astronomy for a general audience; started October 2026

Public Talk on the Hubble Tension, Bulgarian Academy of Sciences 2026
Outreach talk for the general public in Sofia, Bulgaria

Astro Night Public Outreach Talk, University of California, Berkeley 2025
Talk for high school students, community members, and Sequoia Community College students

Physics Fair Volunteer, Johns Hopkins University 2025
Live physics demonstrations to make scientific concepts accessible to the public

Physics Department Outreach Volunteer, Johns Hopkins University 2025
Visited Towson High School to present physics demonstrations and discuss careers in astronomy

Program for Advancing the Health Sciences (PATHS) Mentor 2021–2022
Study habits workshop and twice-monthly one-on-ones with an Alexandria high school student

Application Resources for College and Higher-Education (ARCH) Mentor and Panelist 2021
Scholarship and essay panelist; weekly one-on-ones with an underprivileged college applicant